



Loops: Code That Repeats

Quick facilitation notes & answer key

Goal

Students count, change, and write a repeat loop — learning that the loop runs the blocks inside it a set number of times.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

A block-coding app (Scratch-style) to project; the printed worksheet + a pencil per student.

Run it (about 30 min)

1. Hook: class claps 4 times — say it as a loop, "repeat 4: clap."
2. Project the repeat-4 square loop; count the runs aloud together.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: the repeat number = how many times.

Answer key (checked by the run-gate)

Ex 1 — Bit moves 4 times and turns 4 times.

Ex 2 — Bit moves 6 times (predicted = 6). That is MORE than before.

Ex 3 — repeat 5.

Ex 4 (challenge) — 6 jumps. 2 jumps each time $\times$ 3 repeats = 6.

Watch for & differentiate

Common slip: counting the blocks once instead of once per repeat (esp. Ex 4).

Support: read "repeat 6" as "do it 6 times" and tap each run on fingers.

Extend: in Ex 4, ask what repeat 4×2 jumps would give (8).

Success looks like

The student reads a repeat block to say how many times the inside runs (Ex 1–2), writes a repeat number to hit a target count (Ex 3), and reasons that two inside blocks both run each pass (Ex 4).